Faculty of Information Technology
Capstone Project Academic Year: 2024-2025
Course Code & Name: Database Development with PL/SQL INSY 8311
Lecturer: Eric Maniraguha | eric.maniraguha@auca.ac.rw Date: March 21, 2024

All Groups (Mon, Tues, Wed, Thu)

Instructions: The project will be completed in phases, each with specific requirements. Please keep the following points in mind:

1. **Respect Deadlines:**
   - Deadlines are strict. Late submissions will not be accepted and will result in a score of zero for that phase.
   - Plan and manage your time effectively to ensure timely submissions.
2. **Proposal Submission:**
   - This is a personal project. Late submissions of the proposal will not be accepted and will result in zero points.
3. **Active Participation:**
   - You are expected to fully engage with the project and understand every aspect of it.
   - Ensure you are actively working and contributing to each phase.
4. **Focus on Quality:**
   - Aim for high-quality work in every aspect of the project. Quality will be a key factor in project acceptance.
   - Review your work carefully before submission to avoid last-minute delays.
5. **Exploration of Ideas:**
   - Feel free to explore different ideas during the implementation phase.
   - You may submit multiple ideas, and I will assist you in selecting the best one in the first phase.
6. **Project Acceptance:**
   - Not all projects will be accepted. If your proposal does not meet the required standards, you will be given two additional days to refine or rewrite it.
   - Make sure your proposal is well thought out and clearly defined to avoid this situation.
7. **Report Submission via GitHub:**
   - After the problem statement, you must submit the report for each subsequent phase via GitHub.
   - I will be monitoring all submissions through GitHub, so make sure to keep it updated regularly.

Note: This project is considered your final exam.

Project Outline: PL/SQL Oracle Database Design and Implementation

This project is a comprehensive, multi-phase individual assignment focused on Oracle database design and PL/SQL programming. You will select a complex, real-world problem and develop a working database solution through multiple clearly defined phases.

I. Phase: Problem Statement and Presentation

- **Deadline: March 25, 2025**, at **11:59 PM**
- **Weight:** 5 points

1. Objective

Identify a real-world problem that requires a PL/SQL-based Oracle database solution. The problem should be complex enough to involve **multiple entities, relationships, and business logic**.

2. Problem Statement Guidelines

Your submission should clearly outline:

- **Problem Definition:** What issue are you solving?
- **Context:** Where and how will the system be used? (e.g., hospitals, universities, businesses)
- **Target Users:** Who will benefit from your system?
- **Project Goals:** What outcomes are expected? (e.g., automate workflows, improve accuracy, enhance data security)

2. Presentation Requirements

Prepare a brief in-class presentation that explains:

- The **project objectives**
- The **main entities** in your database
- The **anticipated benefits** of your solution

3. Document Format

Please submit your work as a **PowerPoint presentation** with the following formatting:

- **Font:** Helvetica
- **Use bullet points** throughout the slides
- **Number of slides:** Maximum of **three slides** (you may submit one or two slides only)
- Ensure clarity and consistency in layout

4. File Naming Convention

- Name your file using the following format:
- **Grp_studentID_firstName_PLSQL**

- *Example:* Mon_121212_eric_plsql
- Make sure to use **underscores** (_) between each part.

5. **Submission Method**
    - Email your document to: **eric.maniraguha@auca.ac.rw**
    - *This is an individual assignment.* Each student must submit their own work.
    - Late submissions will not be accepted and will receive **zero points**.

Additional Notes

- **Independent Work:** Each student is expected to work independently and complete all project phases step by step.
- **Creativity & Complexity:** Choose a unique, meaningful problem. Projects that demonstrate originality and thoughtful design will be rewarded.
- **Progress Tracking:** Regular updates and presentations will be required to assess your progress.
- **Use of AI Tools:** AI tools may be used for research and paraphrasing. However, **plagiarism will not be tolerated**. Ensure all content is your original work.
- **Bonus Opportunity:** Exceptional performance during each phase may earn you **2 bonus points**.
- **Upcoming Phases:** Additional project phases will be shared as the course progresses.
- **No Excuses:** Deadlines are strict. No delays will be tolerated unless you're repeating the course. Stay disciplined and focused throughout.

Final Note:
This project serves as your **Final Exam**. Give it your best effort! Your commitment and creativity will define your success.

For I know the plans I have for you," declares the Lord, "plans to prosper you and not to harm you, plans to give you hope and a future."
— Jeremiah 29:11 (NIV)

Good luck!